STA 5635 HW 5

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Problem 1

(1a) Loss function: $L(\boldsymbol{w}) = \frac{1}{N}\sum_{i=1}^{N} f(y_i\boldsymbol{x}_i^T\boldsymbol{w}) + s||\boldsymbol{w}||_2^2$, where $f(x) = \max(0, 1-x)$. I do not believe that the derivative of $f(x)$ exists, at least not cleanly. The portion of the loss consisting of the average of the function only contributes if the function doesn't return 0. In other words:

$$f(y_i\boldsymbol{x}_i^T\boldsymbol{w}) = \begin{cases} 0 & y_i\boldsymbol{x}_i^T\boldsymbol{w} \geq 1 \\ 1 - y_i\boldsymbol{x}_i^T\boldsymbol{w} & y_i\boldsymbol{x}_i^T\boldsymbol{w} < 1 \end{cases}$$

So for a point where $y_i\boldsymbol{x}_i^T\boldsymbol{w} < 1$, the gradient is $\frac{\partial}{\partial \boldsymbol{w}}(1 - y_i\boldsymbol{x}_i^T\boldsymbol{w}) = -y_i\boldsymbol{x}_i^T$. For a points where $y_i\boldsymbol{x}_i^T\boldsymbol{w} \geq 1$, the gradient is $\frac{\partial}{\partial \boldsymbol{w}}(0) = 0$. All together, the total gradient is:

$$\nabla_{\boldsymbol{w}} = \frac{-1}{N}\sum_{i:(y_i\boldsymbol{x}_i^T\boldsymbol{w})<1} (y_i\boldsymbol{x}_i^T) + 2s\boldsymbol{w}$$

We plot the training loss vs. iteration number is table 1.

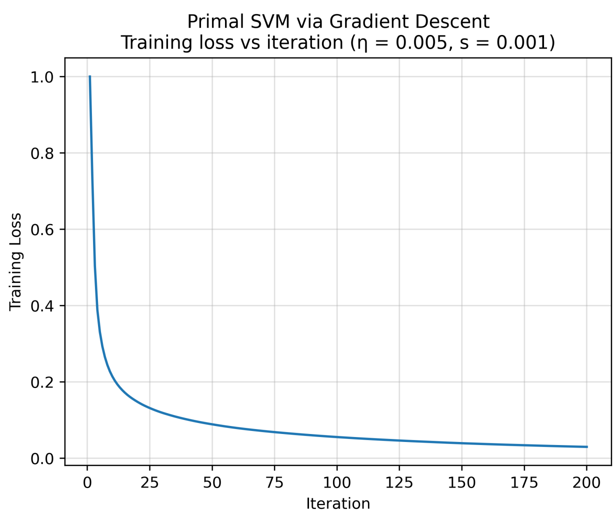

Figure 1: Primal SVM Training Loss vs. Iterations

The table outlining the training and testing misclassification errors is shown in figure 1:

Method	Train Misclass	Test Misclass
Primal SVM	0.006333	0.019000

Table 1: Train and Test Misclassification Errors, Primal SVM via Gradient Descent.

Method	Train Misclass	Test Misclass	Num. Support Vectors
Linear SVM	0.000000	0.019000	1236

Table 2: Train, Test Misclassification Errors and Number of Support Vectors, Linear SVM, C=1

(1b) Use SVM package to train linear SVM, C=1. We can report the training errors and number of support vectors in table 2: Compared to part a, the test misclassification error is the same, but the train misclassification error is slightly lower.

(1c) Use SVM package to train a SVM with polynomial kernel of degree 2. We can report the training error and number of support vectors in table 3: Compared to part b, the

Method	Train Misclass	Test Misclass	Num. Support Vectors
Quadratic SVM	0.000167	0.025000	4744

Table 3: Train, Test Misclassification Errors and Number of Support Vectors, Quadraticc SVM, C=1

number of support vectors is almost four times as great.

(1d) Train an SVM classifer with C=1 and RBF kernels $\gamma = 10^{-i}, i \in \{1, 2, \ldots, 6\}$. We can plot the train and test misclassification errors vs γ in figure 2: We can also plot the number of support vectors vs γ, with a dashed line showing the number of support vectors in part b in figure 3:

(1e) To help with part d, we consider trhe following table 4:

Of all the models considered, it appears as though model a and model b both obtained the lowest test error of 0.019. Of models b through d, the model in part b (quadratic SVM) has the smallest number of support vectors (1236).

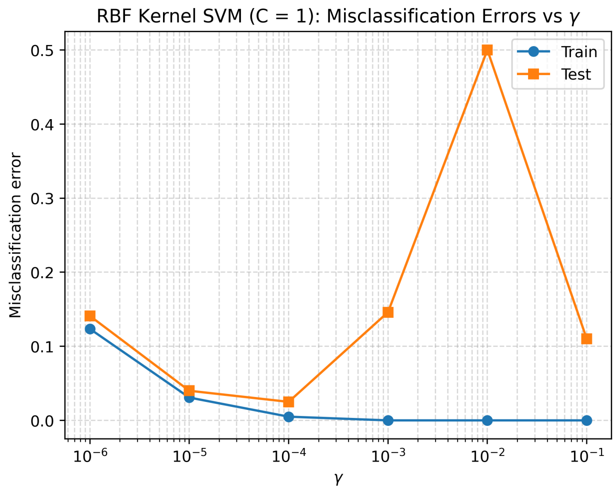

Figure 2: SVM Classifier, C=1, Train and Test Misclassification Errors vs γ.

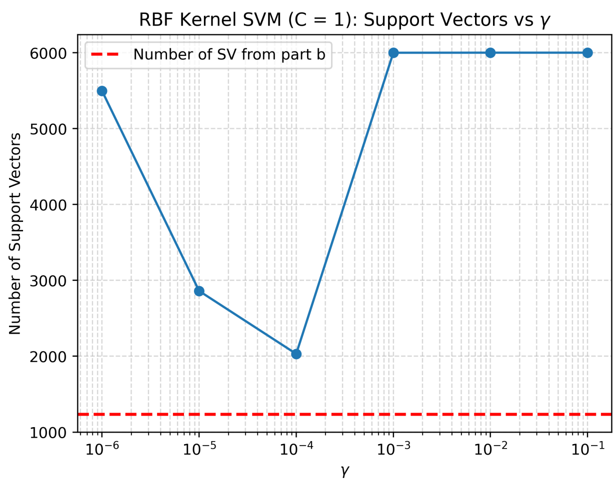

Figure 3: SVM Classifier, C=1, Number of Support Vectors vs γ. Dashed line indicates number of support vectors from part b (quadratic SVM).

γ	Train Error	Test Error	Support Vectors
0.100000	0.000000	0.110000	6000
0.010000	0.000000	0.500000	6000
0.001000	0.000000	0.146000	5999
0.000100	0.005000	0.025000	2031
0.000010	0.030833	0.040000	2860
0.000001	0.123500	0.141000	5497

Table 4: Train, Test Misclassification Errors and Number of Support Vectors, part d

Code